

Rodrigo Alexander Fernandez Huarca

AI Engineer | Backend & Cloud-Native Systems

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PROFESSIONAL SUMMARY

Software Engineer building applied AI systems and the backend infrastructure that runs them in production. Designs RAG pipelines end to end – document ingestion, embeddings, vector retrieval and context grounding – and ships them as containerized FastAPI services on Google Cloud Run. Comfortable across Python, C# (.NET) and TypeScript, and across GCP, Azure and AWS.

section[Technical Skills]Technical Skills

- AI & LLM: RAG architectures, LangChain, prompt orchestration, Gemini API, Gemini Nano (on-device), OpenAI API, FAISS, sentence-transformers embeddings.
- Machine Learning: scikit-learn, HDBSCAN, K-Means, SciBERT, spaCy, TF-IDF, OpenCV, Pandas, NumPy.
- Backend & APIs: Python (FastAPI, Flask, Uvicorn), C# (.NET), TypeScript (NestJS, Node.js), Java (Spring Boot), REST, WebSockets.
- Cloud & DevOps: Google Cloud (Cloud Run), Azure, AWS (S3), Docker (multi-stage), Terraform, GitHub Actions, Grafana, Azure Monitor, Linux.
- Databases: PostgreSQL, PostGIS, FAISS (vector), MongoDB, SQLite.
- Practices: Layered/clean architecture, containerized environments, Git, Agile.
- Languages: Spanish (native), English (B2).

section[Work Experience]Work Experience

Software Engineer — Backend & AI Automation

May 2026 – Present

SimiAI

Remote

- Designed and led the proof of concept for a Python agent engine (FastAPI) with tool-orchestration logic, defining the skill registry and data flow for automated engineering design.
- Built the integration bridge that parses AI-processed field data – photogrammetry, GPS and NLP output – into standardized JSON payloads consumed downstream by the CAD automation pipeline.
- Programmed C# .NET logic that inserts dynamic blocks (poles, conduits, utility boxes) at precise georeferenced coordinates and orientations derived from routing parameters.
- Implemented asynchronous WPF/PaletteSet dockable panels (MVVM) so cloud-synced alerts and field media render without blocking the CAD UI thread.
- *Tech Stack*: C# .NET (AutoCAD API), Python (FastAPI), AWS (S3, SDK for .NET), Azure, Docker, PostgreSQL / PostGIS.

Jr. Software Engineer — Cloud & DevOps (CI/CD)

Nov 2025 – May 2026

Committed Organization

Remote

- Deployed and monitored backend inference services across Azure and GCP, building Grafana and Azure Monitor dashboards to track service health and support incident response.
- Automated cloud infrastructure provisioning with Terraform templates for reproducible, consistent environments.
- Configured Azure virtual networks, storage accounts and access policies; maintained CI/CD pipelines in GitHub Actions.
- *Tech Stack*: Azure, GCP, AWS, Terraform, Docker, Grafana, GitHub Actions, Bash, Linux.

Jr. Software Engineer — Full-Stack

Apr 2025 – Nov 2025

Committed Organization

Remote

- Built and maintained modular backend services in Java and Spring Boot for transactional data flows, containerized with Docker for parity between local and production environments.

– *Tech Stack*: Java, Spring Boot, React, PostgreSQL, Docker, Git.

Machine Learning Engineer — Research Project

Jun 2025 – Dec 2025

Universidad Nacional de San Agustín (UNSA)

Arequipa, Peru

- Led development and training of ML models (Python, scikit-learn) for crop classification and phenological stage detection from satellite imagery, reaching +75% accuracy.
- Built Python/Pandas data pipelines with OpenCV preprocessing for feature extraction, cutting manual analysis time by ~80%.
- Containerized and deployed the inference services on Google Cloud Run.
- *Tech Stack*: Python, FastAPI, Pandas, OpenCV, scikit-learn, Google Earth Engine, Google Cloud Run, Docker.

section[AI Projects]AI Projects

Law Copilot — RAG over Peruvian legislation

Jan 2026

Lead AI & Cloud Engineer — Google AI Partner Catalyst

- Built a specialized vector knowledge base indexing the Political Constitution of Peru, the Civil Code, the Consumer Protection Code and the Law against Violence towards Women, so answers stay grounded in specific statutes.
- Implemented the retrieval layer with FAISS and sentence-transformers embeddings, using dynamic context injection to reduce hallucination on legal questions.
- Deployed the FastAPI backend to Google Cloud Run with multi-stage Docker builds to shrink the image and cut cold-start latency.
- *Tech Stack*: Python (FastAPI), LangChain, FAISS, Gemini API, OpenAI API, Next.js, Google Cloud Run, Docker.

Quechua Q&A — RAG for a low-resource language

2026

AI & Backend Engineer

- Built a question-answering system over Quechua grammar documents, returning answers with the source sections and page numbers used, so every response is traceable.
- Structured the FastAPI backend in layered architecture (API / services / repositories / core) with Pydantic models and async I/O.
- Set up GitHub Actions CI/CD with deployment configurations for both Azure and GCP.
- *Tech Stack*: Python (FastAPI), Gemini API, Pydantic, Next.js, TypeScript, GitHub Actions, Azure, GCP.

On-Device AI Study Assistant

Nov 2025

AI Engineer — Google Chrome Built-in AI Challenge 2025

- Integrated Gemini Nano inside Chrome to generate summaries, flashcards and five types of exercises fully offline – study documents never leave the user's device.
- Split the workload deliberately: local Nano for privacy-sensitive document processing, cloud Gemini via FastAPI and LangChain for learning roadmaps and generated games.
- *Tech Stack*: Next.js 14, TypeScript, Gemini Nano (Prompt API), Gemini API, Python (FastAPI), LangChain, PyPDF2.

Space Biology Knowledge Engine

Oct 2025

AI Engineer — NASA Space Apps Challenge 2025

- Clustered space-biology literature by meaning using SciBERT embeddings and HDBSCAN, with automatic cluster labeling (KeyBERT, TF-IDF) and an agent that extracts research evidence from each paper.
- Exposed the result as an interactive knowledge map backed by a FastAPI service.
- *Tech Stack*: Python (FastAPI), SciBERT, HDBSCAN, scikit-learn, spaCy, transformers, KeyBERT, React.

ONISAT CanSat 2026 — Telemetry & Imaging System

Dec 2025 – Present

Lead Software Engineer

- Owned the full data path for a CanSat mission: C++17 flight software on a Raspberry Pi Zero 2W, a 433 MHz LoRa downlink, and a Python ground station that reassembles fragmented images, recovering packets lost in flight with OpenCV inpainting instead of dropping frames.
- *Tech Stack*: C++17, Raspberry Pi, ESP32-S3, LoRa (SX1278), Python, OpenCV, WebSockets, Next.js, Three.js, Docker.

section[Education]Education Bachelor of Science in Systems Engineering
Universidad Nacional de San Agustín (UNSA)

Expected 2027
Arequipa, Peru